RAFFLES INSTITUTION
H2 Mathematics 9758
2026 Year 6 Term 3 Revision 5 (Zeta)
Topic(s): Vectors
(Vectors II and III (Solutions))

Source of Question: JPJC Prelim 9758/2022/02/Q2

1 **(a)** With reference to the origin O, the points A, B and X are $\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{OB}=\mathbf{b}$ and $\overrightarrow{OX}=\frac{1}{8}\mathbf{a}+\frac{3}{8}\mathbf{b}$. The point Y lies on AB such that O, X and Y are collinear. Express $\overrightarrow{OY}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ and find the ratio of $AY:YB$. [5]

(b) The points P, Q and R have non zero position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ respectively. P and Q are fixed, and R varies. Describe geometrically the set of possible positions of the point R such that

(i) $(\mathbf{r}-\mathbf{p})\times\mathbf{q}=\mathbf{0}$, [2]

(ii) $(\mathbf{r}-\mathbf{p}).\mathbf{q}=0$. [2]

Solution:

(a)

Y lies on line $AB \Rightarrow \overrightarrow{OY}=\mathbf{a}+\beta(\mathbf{b}-\mathbf{a})$, for some $\beta\in\mathbb{R}$.

Since O, X and Y are collinear,

$\mathbf{a}+\beta(\mathbf{b}-\mathbf{a})=\alpha\left(\frac{1}{8}\mathbf{a}+\frac{3}{8}\mathbf{b}\right)$ for some $\alpha\in\mathbb{R}$.

Since $\mathbf{a}$ and $\mathbf{b}$ are non-parallel and nonzero,

Comparing,

$1-\beta=\frac{\alpha}{8}$ ---(1)

$\beta=\frac{3\alpha}{8}$ ---(2)

Solving (1) and (2), $\alpha=2$, $\beta=\frac{3}{4}$

$\overrightarrow{OY}=2\left(\frac{1}{8}\mathbf{a}+\frac{3}{8}\mathbf{b}\right)=\frac{1}{4}\mathbf{a}+\frac{3}{4}\mathbf{b}$

$\overrightarrow{AY}=\frac{3}{4}\overrightarrow{AB}$

$AY:YB=3:1$

(b)(i)	$(\mathbf{r}-\mathbf{p})\times\mathbf{q}=\mathbf{0}$ We see that either $(\mathbf{r}-\mathbf{p})//\mathbf{q}$ or $(\mathbf{r}-\mathbf{p})=\mathbf{0}$ $\mathbf{r}=\mathbf{p}+k\mathbf{q},\ k\neq 0$ or $\mathbf{r}=\mathbf{p}$ Thus $\mathbf{r}=\mathbf{p}+k\mathbf{q},\ k\in\mathbb{R}$ The set of all possible positions of the point R is the line that passes through the point P and parallel to the vector $\mathbf{q}$.
(b)(ii)	$(\mathbf{r}-\mathbf{p}).\mathbf{q}=0$ $\mathbf{r}.\mathbf{q}-\mathbf{p}.\mathbf{q}=0$ $\mathbf{r}.\mathbf{q}=\mathbf{p}.\mathbf{q}$ The set of all possible positions of the point R is the plane that is perpendicular to $\mathbf{q}$ and containing the point P.

Source of Question: NJC Prelim 9758/2025/P2/Q3

2. With respect to an origin O, points P and Q have variable position vectors $\mathbf{p}$ and $\mathbf{q}$ respectively, given by $\mathbf{p}=(\cos t)\mathbf{i}+(\sin t)\mathbf{j}-\mathbf{k}$, $\mathbf{q}=(\cos 2t)\mathbf{i}-(\sin 2t)\mathbf{j}+\alpha\mathbf{k}$, where α is a real parameter and $0<t<\pi$.

(i) By considering the cross product or otherwise, find the exact values of t and α if O, P and Q are collinear. [5]

It is given that $\alpha = 0.5$.

(ii) Show that $\mathbf{p}\cdot\mathbf{q}=\cos 3t-0.5$. Find the greatest length of projection of $\mathbf{q}$ onto $\mathbf{p}$. [4]

Solution:

(i)

Method 1: Use cross product

$$\mathbf{p}=\begin{pmatrix}\cos t\\ \sin t\\ -1\end{pmatrix},\ \mathbf{q}=\begin{pmatrix}\cos 2t\\ -\sin 2t\\ \alpha\end{pmatrix}$$

$$\mathbf{p}\times\mathbf{q}=\begin{pmatrix}\cos t\\ \sin t\\ -1\end{pmatrix}\times\begin{pmatrix}\cos 2t\\ -\sin 2t\\ \alpha\end{pmatrix}$$

$$=\begin{pmatrix}\alpha\sin t-\sin 2t\\ -\alpha\cos t-\cos 2t\\ -\sin 2t\cos t-\cos 2t\sin t\end{pmatrix}$$

$$=\begin{pmatrix}0\\0\\0\end{pmatrix}$$

Equate **k**-component to 0

$$-\sin 2t\cos t-\cos 2t\sin t=0$$
$$-(\sin 2t\cos t+\cos 2t\sin t)=0$$
$$-\sin 3t=0$$

For $-\sin 3t=0\Rightarrow t=\frac{\pi}{3},\frac{2\pi}{3}$.

Equate **i**-component to 0

$$\alpha=\frac{\sin 2t}{\sin t}=2\cos t \quad (\text{note: } \sin t\neq 0)$$

When $t=\frac{\pi}{3}, \alpha=1$

When $t=\frac{2\pi}{3}, \alpha=-1$

Method 2: **q** is a scalar multiple of **p**

$$\mathbf{q}=k\mathbf{p}\Rightarrow\begin{pmatrix}\cos 2t\\ -\sin 2t\\ \alpha\end{pmatrix}=k\begin{pmatrix}\cos t\\ \sin t\\ -1\end{pmatrix}$$

$$\begin{pmatrix}\cos 2t\\ -\sin 2t\\ \alpha\end{pmatrix}=\begin{pmatrix}k\cos t\\ k\sin t\\ -k\end{pmatrix}$$

From **j**-component:

$$-\sin 2t=k\sin t$$
$$-2\cos t\sin t=k\sin t$$
$$-2\cos t\sin t-k\sin t=0$$
$$\sin t(-2\cos t-k)=0$$
$$\sin t=0 \text{ or } -2\cos t-k=0$$
$$t=0\ (\text{rejected } \because 0<t<\pi) \text{ or } \cos t=-\frac{k}{2}$$

From **i**-component:

$$\cos 2t=k\cos t$$
$$2\cos^2 t-1-k\cos t=0$$
$$2\cos^2 t-k\cos t-1=0$$

Sub $\cos t=-\frac{k}{2}$ into equation $2\cos^2 t-k\cos t-1=0$:

	Note: Same corresponding values of α are obtained when we equate **j**-component to 0 Final answers: $t=\frac{\pi}{3}, \alpha=1;$ $t=\frac{2\pi}{3}, \alpha=-1$	$2\left(-\frac{k}{2}\right)^2-k\left(-\frac{k}{2}\right)-1=0$ $\frac{k^2}{2}+\frac{k^2}{2}=1$ $k=\pm 1$ Since $\alpha=-k$, $\alpha=-1$ or $\alpha=1$. When $\alpha=1$, $t=\cos^{-1}\left(\frac{1}{2}\right)=\frac{\pi}{3}$ When $\alpha=-1$, $t=\cos^{-1}\left(-\frac{1}{2}\right)=\frac{2\pi}{3}$
(ii)	$\mathbf{p}\cdot\mathbf{q}$ $=\begin{pmatrix}\cos t\\ \sin t\\ -1\end{pmatrix}\cdot\begin{pmatrix}\cos 2t\\ -\sin 2t\\ 0.5\end{pmatrix}$ $=\cos t\cos 2t-\sin t\sin 2t-0.5$ $=\cos(2t+t)-0.5$ $=\cos 3t-0.5$ $\frac{\lvert\mathbf{p}\cdot\mathbf{q}\rvert}{\lvert\mathbf{p}\rvert}=\frac{\lvert\cos 3t-0.5\rvert}{\sqrt{2}}$ $=\frac{1}{\sqrt{2}}\lvert\cos 3t-0.5\rvert$ Since $0<t<\pi$, greatest length $=\frac{1}{\sqrt{2}}\lvert-1-0.5\rvert$ $=\frac{3}{2\sqrt{2}}$ or 1.06 (3 s.f.)	

Source of Question: HCI Prelim 9758/2023/02/Q5

3 The equations of two planes p_1 and p_2 are

$$2x + sy + z = -3 \ ,$$
$$x - 2y + 3z = 1 \ ,$$

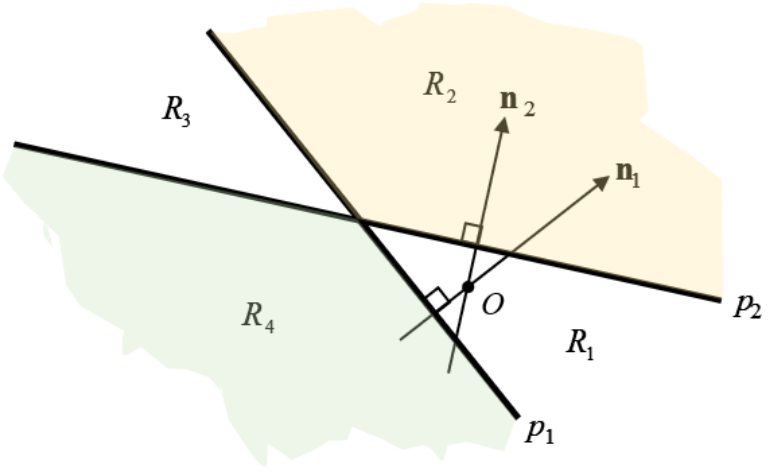
respectively, where s is a negative real constant.

(a) Find a vector parallel to both p_1 and p_2, giving your answer in terms of s. [2]

(b) It is given that p_1 and p_2 intersect in a line l, and l meets the xz-plane at a point A. Find the coordinates of A. [3]

(c) The shortest distance between the origin O and p_1 is $\frac{3}{\sqrt{6}}$. Find the value of s. [2]

It is now given that $s = -2$, with the normal vectors to p_1 and p_2 denoted as $\mathbf{n_1} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}$ and $\mathbf{n_2} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ respectively. It is given that p_1 and p_2 divide the real space into four regions R_1, R_2, R_3 and R_4 as shown in the following two-dimensional diagram (not drawn to scale).



(d) Relative to the origin O, the point B has position vector $\mathbf{k}$. Determine whether B lies in R_1, R_2, R_3 or R_4, justifying your answer. [2]

Solution:

(a)

A vector parallel to both p_1 and p_2 is $\begin{pmatrix}2\\s\\1\end{pmatrix}\times\begin{pmatrix}1\\-2\\3\end{pmatrix}=\begin{pmatrix}3s+2\\-5\\-4-s\end{pmatrix}$

(b)

xz-plane $\Rightarrow\ y=0$

$2x+z=-3$ ...(1)

$x+3z=1$ ...(2)

$2\times(2)-(1)\text{: } 5z=5\ \Rightarrow\ z=1$

$\therefore\ x=-2$

Hence coordinates of A is $(-2,0,1)$.

(c)

$p_1:\ \underset{\sim}{r}.\begin{pmatrix}2\\s\\1\end{pmatrix}=-3$

$$\underset{\sim}{r}.\frac{1}{\sqrt{2^2+s^2+1^2}}\begin{pmatrix}2\\s\\1\end{pmatrix}=\frac{1}{\sqrt{2^2+s^2+1^2}}(-3)=\frac{-3}{\sqrt{s^2+5}}$$

$\therefore$ shortest distance between O and p_1 is

$$\left|\frac{-3}{\sqrt{s^2+5}}\right|=\frac{3}{\sqrt{6}}$$

$$\frac{3}{\sqrt{s^2+5}}=\frac{3}{\sqrt{6}}$$

$$\sqrt{\frac{6}{s^2+5}}=1$$

$$\frac{6}{s^2+5}=1$$

$$6=s^2+5$$

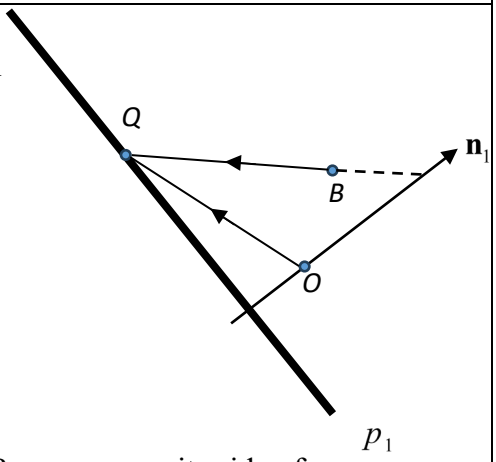
$$s^2=1$$

$$s=\pm1 \quad \text{(reject } s=1 \text{ since } s<0\text{)}$$

$\therefore\ s=-1$

(d)

Given $\overrightarrow{OB}=\underset{\sim}{k}=\begin{pmatrix}0\\0\\1\end{pmatrix}$, $p_1:\ \underset{\sim}{r}.\begin{pmatrix}2\\-2\\1\end{pmatrix}=-3$, $p_2:\ \underset{\sim}{r}.\begin{pmatrix}1\\-2\\3\end{pmatrix}=1$



<u>Method 1:</u>

Consider the origin O:

$p_1:\ 2(0)-2(0)+(0)=0>-3$

$p_2:(0)-2(0)+3(0)=0<1$

Consider the point B:

$p_1:\ 2(0)-2(0)+(1)=1>-3$

$p_2:(0)-2(0)+3(1)=3>1$

Hence, O and B are on the same side of p_1 but O and B are on opposite side of p_2.

$\therefore\ B$ is in R_2.

Method 2: Using angles

$p_1 : \underset{\sim}{r} . \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = -3$

$Q(0,0,-3)$ is a point that lies in p_1.

Since $\overrightarrow{OQ} \cdot \mathbf{n}_1 = -3 < 0$,

then the angle between OQ and $\mathbf{n}_1$ is obtuse.

$\overrightarrow{BQ} \cdot \mathbf{n}_1 = \begin{pmatrix} 0 \\ 0 \\ -4 \end{pmatrix} . \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = -4 < 0$,

then the angle between BQ and $\mathbf{n}_1$ is obtuse.

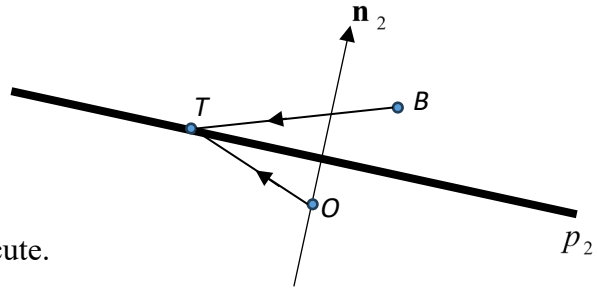
$\therefore$ O and B are on same side of p_1. $---(1)$

$p_2 : \underset{\sim}{r} . \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = 1$

$T(1,0,0)$ is a point that lies in $\mathbf{n}_2$.

Since $\overrightarrow{OT} \cdot \mathbf{n}_2 = 1 > 0$,

then the angle between OT and $\mathbf{n}_2$ is acute.



$\overrightarrow{BT} \cdot \mathbf{n}_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} = -2 < 0$,

then the angle between BT and $\mathbf{n}_2$ is obtuse.

$\therefore$ O and B are on opposite side of p_2. $---(2)$

Combining (1) and (2) with reference to given diagram, $\therefore$ B lies in R_2.

Method 3: (using shortest distance of planes to origin O)

Let p_3 be plane containing B and parallel to p_1.

Equation of p_3 is $\underset{\sim}{r} . \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} . \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = 1$

$$\underset{\sim}{r} . \frac{1}{\sqrt{2^2 + (-2)^2 + 1^2}} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2^2 + (-2)^2 + 1^2}}$$

i.e. $\underset{\sim}{r} . \frac{1}{3} \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} = \frac{1}{3} > 0$

Since equation of p_1 $\underset{\sim}{r}.\frac{1}{3}\begin{pmatrix}2\\-2\\1\end{pmatrix}=\frac{-3}{3}=-1<0$,

$\therefore$ p_1 and p_3 are on opposite sides of origin O. …(1)

Let p_4 be plane containing B and parallel to p_2.

Equation of p_4 is $\underset{\sim}{r}.\begin{pmatrix}1\\-2\\3\end{pmatrix}=\begin{pmatrix}0\\0\\1\end{pmatrix}.\begin{pmatrix}1\\-2\\3\end{pmatrix}=3$

$$r.\frac{1}{\sqrt{1^2+(-2)^2+3^2}}\begin{pmatrix}1\\-2\\3\end{pmatrix}=\frac{3}{\sqrt{1^2+(-2)^2+3^2}}=\frac{3}{\sqrt{14}}>0$$

i.e. $r.\frac{1}{\sqrt{14}}\begin{pmatrix}1\\-2\\3\end{pmatrix}=\frac{3}{\sqrt{14}}>0$

Since equation of p_2 $\underset{\sim}{r}.\frac{1}{\sqrt{14}}\begin{pmatrix}1\\-2\\3\end{pmatrix}=\frac{1}{\sqrt{14}}>0$,

$\therefore$ p_2 and p_4 are on same side of origin O. …(2)

Combining (1) and (2) with reference to given diagram, $\therefore$ B lies in R_2.

Source of Question: NYJC Prelim 9758/2022/01/Q10

4 Two charged particles, U and V, are confined to the planes F_1 and F_2 with position vectors given by

$$(3+6p)\mathbf{i}+(1+4p+q)\mathbf{j}+(6+2p-4q)\mathbf{k} \text{ and } (-9+3p)\mathbf{i}+(1+p-2q)\mathbf{j}+(3-p+8q)\mathbf{k}$$

respectively, where $p,q\in\mathbb{R}$.

(i) Obtain the equation of F_1 in scalar product form. [2]

(ii) Find the acute angle between F_1 and F_2. [3]

The forces of the two particles U and V allow another charged particle W to remain suspended between them, such that $2UW = WV$.

(iii) As the positions of U and V vary, show that the set of points described by the path of W is a line l, whose vector equation is to be determined. [3]

(iv) An uncharged particle A is fired along a path described by a line $\mathbf{r} = s\mathbf{k}$, where $s\in\mathbb{R}$ and crosses F_1 at some instant in time. Find the shortest distance of A from l at this instant. [4]

Solution:

(i)

$$F_1:\ \mathbf{r}=\begin{pmatrix}3+6p\\1+4p+q\\6+2p-4q\end{pmatrix}=\begin{pmatrix}3\\1\\6\end{pmatrix}+p\begin{pmatrix}6\\4\\2\end{pmatrix}+q\begin{pmatrix}0\\1\\-4\end{pmatrix}$$

$$\mathbf{n}_1=\begin{pmatrix}6\\4\\2\end{pmatrix}\times\begin{pmatrix}0\\1\\-4\end{pmatrix}=\begin{pmatrix}-18\\24\\6\end{pmatrix}=6\begin{pmatrix}-3\\4\\1\end{pmatrix}$$

$$F_1:\mathbf{r}\cdot\begin{pmatrix}-3\\4\\1\end{pmatrix}=\begin{pmatrix}3\\1\\6\end{pmatrix}\cdot\begin{pmatrix}-3\\4\\1\end{pmatrix}=1$$

$$\mathbf{r}\cdot\begin{pmatrix}-3\\4\\1\end{pmatrix}=1$$

(ii)

$$F_2:\ \mathbf{r}=\begin{pmatrix}-9+3p\\1+p-2q\\3-p+8q\end{pmatrix}=\begin{pmatrix}-9\\1\\3\end{pmatrix}+p\begin{pmatrix}3\\1\\-1\end{pmatrix}+q\begin{pmatrix}0\\-2\\8\end{pmatrix}$$

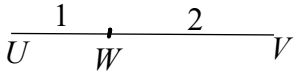
$$\mathbf{n}_2=\begin{pmatrix}3\\1\\-1\end{pmatrix}\times\begin{pmatrix}0\\-2\\8\end{pmatrix}=\begin{pmatrix}6\\-24\\-6\end{pmatrix}=-6\begin{pmatrix}-1\\4\\1\end{pmatrix}$$

$$\theta=\cos^{-1}\frac{\left|\begin{pmatrix}-3\\4\\1\end{pmatrix}\cdot\begin{pmatrix}-1\\4\\1\end{pmatrix}\right|}{\sqrt{9+16+1}\sqrt{1+16+1}}=\cos^{-1}\left|\frac{20}{\sqrt{26}\sqrt{18}}\right|\approx 22.4^0$$

(iii)

$$\overrightarrow{OU}=\begin{pmatrix}3+6p\\1+4p+q\\6+2p-4q\end{pmatrix}\qquad\overrightarrow{OV}=\begin{pmatrix}-9+3p\\1+p-2q\\3-p+8q\end{pmatrix}$$

Given $2UW=WV$ (or equivalently $\frac{UW}{WV}=\frac{1}{2}$). We assume that U, W, V are collinear.



$$\overrightarrow{OW}=\frac{1}{3}\left(\overrightarrow{OV}+2\overrightarrow{OU}\right)$$

$$= \frac{1}{3}\left[\begin{pmatrix} -9+3p \\ 1+p-2q \\ 3-p+8q \end{pmatrix} + 2\begin{pmatrix} 3+6p \\ 1+4p+q \\ 6+2p-4q \end{pmatrix}\right]$$

$$= \frac{1}{3}\begin{pmatrix} -3+15p \\ 3+9p \\ 15+3p \end{pmatrix} = \begin{pmatrix} -1+5p \\ 1+3p \\ 5+1p \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix} + p\begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}$$

$\therefore$ Path of W is a line l with equation $\mathbf{r} = \begin{pmatrix} -1 \\ 1 \\ 5 \end{pmatrix} + p\begin{pmatrix} 5 \\ 3 \\ 1 \end{pmatrix}, p \in \mathbb{R}$

(iv)

Method 1: Use the scalar product form of F_1

Since point A is on F_1 and on the line $\mathbf{r} = s\mathbf{k} = s\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$,

we have $s\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix} = 1$.

Therefore, $s = 1$ and $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Method 2: Use the parametric form of F_1

Since point A is on F_1 and on the line $\mathbf{r} = s\mathbf{k} = s\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$,we have

$$\mathbf{s}\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 6 \end{pmatrix} + p\begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix} + q\begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} ---(*)$$

$3+6p=0$ and $1+4p+q=0$

$p = -\frac{1}{2}$ and $1+4(-\frac{1}{2})+q=0$

$q = 1$

Sub back (*), $\overrightarrow{OA} = \begin{pmatrix} 3 \\ 1 \\ 6 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 6 \\ 4 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ -4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

For the perpendicular distance between A and line l

Method 1: Use length of projection formula

$$\text{Distance of } A \text{ to } l = \frac{\left|\left(\begin{pmatrix}-1\\1\\5\end{pmatrix}-\begin{pmatrix}0\\0\\1\end{pmatrix}\right)\times\begin{pmatrix}5\\3\\1\end{pmatrix}\right|}{\sqrt{25+9+1}} = \frac{\left|\begin{pmatrix}-1\\1\\4\end{pmatrix}\times\begin{pmatrix}5\\3\\1\end{pmatrix}\right|}{\sqrt{35}}$$

$$= \frac{\left|\begin{pmatrix}-11\\21\\-8\end{pmatrix}\right|}{\sqrt{35}} = \frac{\sqrt{626}}{\sqrt{35}} = 4.23 \text{ (to 3 s.f.)}$$

Method 2: Find the distance between A and its foot of perpendicular to l (*not recommended*)

Let P be the foot of perpendicular from A to l

$$\overrightarrow{OP} = \begin{pmatrix}-1\\1\\5\end{pmatrix} + p\begin{pmatrix}5\\3\\1\end{pmatrix}, \text{ for some } p \in \mathbb{R}, \text{ and } \overrightarrow{AP} = \begin{pmatrix}-1\\1\\5\end{pmatrix} + p\begin{pmatrix}5\\3\\1\end{pmatrix} - \begin{pmatrix}0\\0\\1\end{pmatrix} = \begin{pmatrix}-1\\1\\4\end{pmatrix} + p\begin{pmatrix}5\\3\\1\end{pmatrix}$$

Since $\overrightarrow{AP}$ is perpendicular to $\begin{pmatrix}5\\3\\1\end{pmatrix}$, $\left[\begin{pmatrix}-1\\1\\4\end{pmatrix} + p\begin{pmatrix}5\\3\\1\end{pmatrix}\right] \bullet \begin{pmatrix}5\\3\\1\end{pmatrix} = 0$

$$2 + (25+9+1)p = 0$$

$$p = -\frac{2}{35}$$

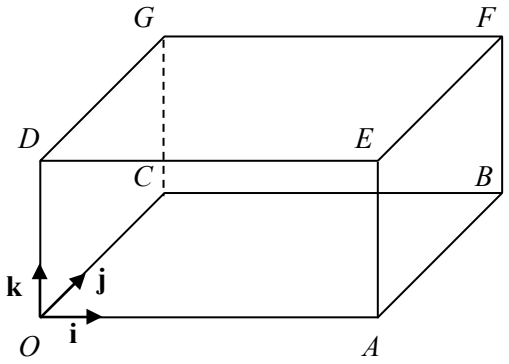
$$\overrightarrow{AP} = \begin{pmatrix}-1\\1\\4\end{pmatrix} - \frac{2}{35}\begin{pmatrix}5\\3\\1\end{pmatrix} = \frac{1}{35}\begin{pmatrix}-45\\29\\138\end{pmatrix}$$

$$\text{Distance of } A \text{ to } l = \left|\overrightarrow{AP}\right| = \frac{\sqrt{(-45)^2 + (29)^2 + (138)^2}}{35} \approx 4.23$$

Source of Question: NYJC CT2 9758/2023/02/Q3

5 A refinery installs pipelines to transport natural gas from one part of the refinery to another. In one section of the refinery, there is a rectangular frame structure as shown in the diagram below. The point O is at one corner of the structure and is taken as the origin for the position vectors. Perpendicular unit vectors **i**, **j** and **k** are parallel to OA, OC and OD respectively. The lengths of OA, OC and OD are 15 units, 10 units and 4 units respectively.

The first pipe enters the lower corner of the structure at O and exits at the diametrically opposed corner F. The second pipe enters at the lower corner A and exits at the opposite lower corner C. The pipes are laid in straight lines and the widths of the pipes can be neglected.



(a) Find the acute angle between the two pipes. [3]

Engineers at the refinery have determined a need to install a support strut between the two pipes to reduce damaging vibrations. To minimise cost, the plan is to install this strut by connecting the closest points between these two skew pipes.

(b) Find a vector perpendicular to both the pipes. It is also known that point P is on the first pipe and point Q is on the second pipe such that PQ is perpendicular to both pipes. Hence, by finding the position vectors of points P and Q, determine the length of the strut in order to minimise cost. [4]

(c) A solar panel $-2x+3y+\beta z=k\beta$, where k and β are constants, is to be installed along DF. For the solar panel to get their best performance in producing the maximum energy, the tilt of the solar panel should be at an angle of 60° with respect to the roof $DEFG$ of the structure. Find the exact values of k and β. [3]

Solution:

(a)

$\overrightarrow{OA}=\begin{pmatrix}15\\0\\0\end{pmatrix}$, $\overrightarrow{OC}=\begin{pmatrix}0\\10\\0\end{pmatrix}$, $\overrightarrow{OF}=\begin{pmatrix}15\\10\\4\end{pmatrix}$ $\quad\overrightarrow{AC}=\begin{pmatrix}0\\10\\0\end{pmatrix}-\begin{pmatrix}15\\0\\0\end{pmatrix}=\begin{pmatrix}-15\\10\\0\end{pmatrix}=5\begin{pmatrix}-3\\2\\0\end{pmatrix}$

Line OF : $\mathbf{r}=\lambda\begin{pmatrix}15\\10\\4\end{pmatrix},\lambda\in\mathbb{R}$

Line AC : $\mathbf{r}=\begin{pmatrix}15\\0\\0\end{pmatrix}+\mu\begin{pmatrix}-3\\2\\0\end{pmatrix},\lambda\in\mathbb{R}$

Let the acute angle between line OF and line AC be θ .

$$\cos\theta=\frac{\left|\begin{pmatrix}15\\10\\4\end{pmatrix}\bullet\begin{pmatrix}-3\\2\\0\end{pmatrix}\right|}{\left|\begin{pmatrix}15\\10\\4\end{pmatrix}\right|\left|\begin{pmatrix}-3\\2\\0\end{pmatrix}\right|}=\frac{25}{\sqrt{341}\sqrt{13}}$$

$\theta=67.9^\circ$ (1 dp)

(b)

$$\begin{pmatrix}15\\10\\4\end{pmatrix}\times\begin{pmatrix}-3\\2\\0\end{pmatrix}=\begin{pmatrix}-8\\-12\\60\end{pmatrix}=-4\begin{pmatrix}2\\3\\-15\end{pmatrix}$$

A vector perpendicular to both pipes is $\begin{pmatrix}2\\3\\-15\end{pmatrix}$.

$\overrightarrow{OP}=\lambda\begin{pmatrix}15\\10\\4\end{pmatrix}$, $\overrightarrow{OQ}=\begin{pmatrix}15\\0\\0\end{pmatrix}+\mu\begin{pmatrix}-3\\2\\0\end{pmatrix}$ for some $\lambda,\mu\in\mathbb{R}$

$$\overrightarrow{PQ}=\begin{pmatrix}15\\0\\0\end{pmatrix}+\mu\begin{pmatrix}-3\\2\\0\end{pmatrix}-\lambda\begin{pmatrix}15\\10\\4\end{pmatrix}$$

Since $\overrightarrow{PQ}$ is parallel to $\begin{pmatrix}2\\3\\-15\end{pmatrix}$,

$$m\begin{pmatrix}2\\3\\-15\end{pmatrix}=\begin{pmatrix}15\\0\\0\end{pmatrix}+\mu\begin{pmatrix}-3\\2\\0\end{pmatrix}-\lambda\begin{pmatrix}15\\10\\4\end{pmatrix},\text{ for some } m\in\mathbb{R}$$

$2m=15-3\mu-15\lambda\Rightarrow 15\lambda+3\mu+2m=15$

$3m=2\mu-10\lambda\Rightarrow 10\lambda-2\mu+3m=0$

$-15m=-4\lambda\Rightarrow 4\lambda-15m=0$

From GC,

	$\lambda = \frac{225}{476}, \mu = \frac{1215}{476}, m = \frac{15}{119}$ $\overrightarrow{OP} = \frac{225}{476}\begin{pmatrix}15\\10\\4\end{pmatrix}$ $\overrightarrow{OQ} = \begin{pmatrix}15\\0\\0\end{pmatrix} + \frac{1215}{476}\begin{pmatrix}-3\\2\\0\end{pmatrix} = \begin{pmatrix}\frac{3495}{476}\\\frac{1215}{238}\\0\end{pmatrix}$ Length of struts $= \left\lvert \frac{15}{119}\begin{pmatrix}2\\3\\-15\end{pmatrix} \right\rvert = \frac{15}{119}\sqrt{238} = 1.9446 = 1.94$ (to 3 s.f.)
(c)	Clearly $\beta \neq 0$. Now since $D(0,0,4)$ lies on the solar panel, $-2(0)+3(0)+\beta(4) = k\beta$ Hence, $k = 4$. $\cos 60^\circ = \left\lvert \frac{\begin{pmatrix}0\\0\\1\end{pmatrix} \bullet \begin{pmatrix}-2\\3\\\beta\end{pmatrix}}{\sqrt{1}\sqrt{13+\beta^2}} \right\rvert$ $\frac{1}{2} = \left\lvert \frac{\beta}{\sqrt{13+\beta^2}} \right\rvert$ $\beta^2 + 13 = 4\beta^2$ $3\beta^2 = 13$ $\beta = \pm\sqrt{\frac{13}{3}}$

Source of Question: HCI CT2 9758/2023/01/Q6

6 The line l has vector equation

$$\mathbf{r} = \begin{pmatrix} b \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ a \\ -5 \end{pmatrix},$$

where λ is a real parameter and a, b are real constants.

The equation of plane π_1 is given as $3x - y + 2z = 7$.

(a) Given that the distance between l and π_1 is non-zero, find the value of a, and the set of values of b. [4]

For the rest of the question, let a be the value found in part **(a)**, and $b = 2$.

(b) Find the vector equation of the line l', where l' is the reflection of l in π_1. [4]

The equation of plane π_2 is given as $-3x + y - 2z = 2$.

(c) Given that the ratio of the distance between π_1 and π_2 to the distance between π_2 and plane π_3 is $2:3$, find the possible cartesian equations of π_3. [4]

Solution:

(a)	Distance between l and π_1 is non-zero $\Rightarrow$ l and π_1 are parallel to each other, and l does not lie on π_1. $\pi_1 : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7$ l and π_1 are parallel to each other $\Rightarrow$ l is perpendicular to the normal of π_1 $\begin{pmatrix} 2 \\ a \\ -5 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 0$ $6 - a - 10 = 0$ $a = -4$ l does not lie on π_1 $\Rightarrow$ the point $(b, -1, 1)$ on l does not lie on π_1

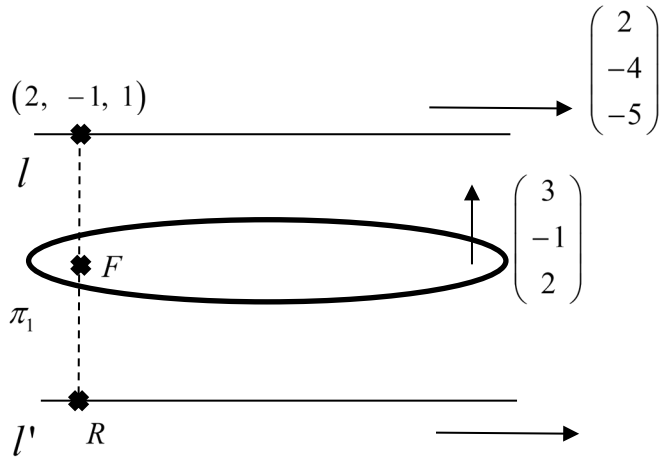
$\begin{pmatrix} b \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} \neq 7$

$3b + 1 + 2 \neq 7$

$b \neq \frac{4}{3}$

The set of values of b is $\left\{ b \in \mathbb{R} \middle| b \neq \frac{4}{3} \right\}$

(b) $a = -4$ implies that l and π_1 are parallel to each other



The direction vector of l' is parallel to l, i.e. a direction vector of l' is $\begin{pmatrix} 2 \\ -4 \\ -5 \end{pmatrix}$.

Let F be the foot of perpendicular of point $(2, -1, 1)$ to π_1

$\overrightarrow{OF} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ for some value of μ

Since F is on π_1, $\overrightarrow{OF} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7 \Rightarrow \begin{pmatrix} 2+3\mu \\ -1-\mu \\ 1+2\mu \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7$

$\Rightarrow 6 + 9\mu + 1 + \mu + 2 + 4\mu = 7$

$\Rightarrow 14\mu = -2$

$\Rightarrow \mu = -\frac{1}{7}$

Therefore $\overrightarrow{OF} = \begin{pmatrix} 2+3\left(-\frac{1}{7}\right) \\ -1-\left(-\frac{1}{7}\right) \\ 1+2\left(-\frac{1}{7}\right) \end{pmatrix} = \begin{pmatrix} \frac{11}{7} \\ -\frac{6}{7} \\ \frac{5}{7} \end{pmatrix}$

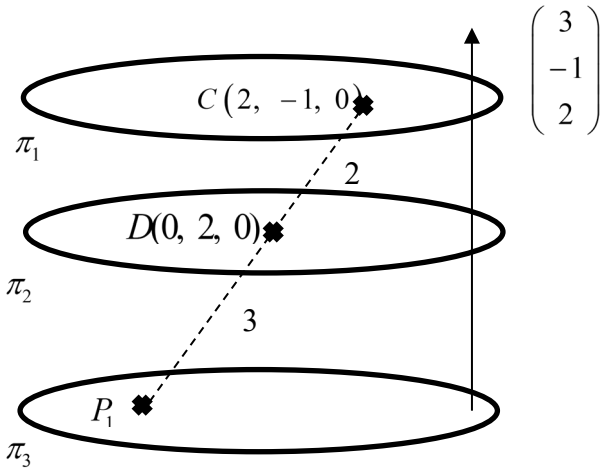
	Let R be the reflection of $(2,\ -1,\ 1)$ about π_1 Using ratio theorem, $\overrightarrow{OF} = \dfrac{\overrightarrow{OR} + \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}}{2} \Rightarrow \overrightarrow{OR} = 2\begin{pmatrix} \frac{11}{7} \\ -\frac{6}{7} \\ \frac{5}{7} \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{8}{7} \\ -\frac{5}{7} \\ \frac{3}{7} \end{pmatrix}$ Equation of line l': $\mathbf{r} = \begin{pmatrix} \frac{8}{7} \\ -\frac{5}{7} \\ \frac{3}{7} \end{pmatrix} + \gamma \begin{pmatrix} 2 \\ -4 \\ -5 \end{pmatrix}$, $\gamma \in \mathbb{R}$
(c)	**Method 1:** Use the formula $\dfrac{\lvert d_2 - d_1 \rvert}{\lvert \mathbf{n} \rvert}$ (refer to Lecture notes C2C Example 5) First, note that π_3 is parallel to π_1, hence in scalar product form, $\pi_3 : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = k$ where k is a constant we have to determine. Now in scalar product form, we have $\pi_1 : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = 7$ and $\pi_2 : \mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = -2$ Distance between π_2 and π_3 is $\dfrac{\lvert k - (-2) \rvert}{\sqrt{14}} = \dfrac{3}{2}\left(\dfrac{\lvert 7 - (-2) \rvert}{\sqrt{14}} \right) = \dfrac{13.5}{\sqrt{14}}$ $\lvert k + 2 \rvert = 13.5$ $k = 11.5$ or -15.5 Hence the two possible cartesian equations π_3 are: $3x - y + 2z = 11.5$ $3x - y + 2z = -15.5$

Method 2: Use Ratio Theorem

Since there is a non-zero distance between π_2 and plane π_3, π_3 is parallel to π_2 (and π_1).

Hence a normal to π_3 is $\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$

$C(2, -1, 0)$ is on π_1,
and $D(0, 2, 0)$ is on π_2



Case 1: π_2 is between π_1 and π_3
Let P_1 be a point on π_3 such that
$CD : DP_1 = 2 : 3$
Using ratio theorem,

$$\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \frac{2\overrightarrow{OP_1} + 3\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}}{5}$$

$$\Rightarrow 2\overrightarrow{OP_1} = 5\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} - 3\begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \Rightarrow \overrightarrow{OP_1} = \begin{pmatrix} -3 \\ \frac{13}{2} \\ 0 \end{pmatrix}$$

Equation of π_3: $\mathbf{r} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ \frac{13}{2} \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} = -9 - \frac{13}{2} + 0 = -\frac{31}{2}$

Cartesian equation of π_3: $3x - y + 2z = -\frac{31}{2}$

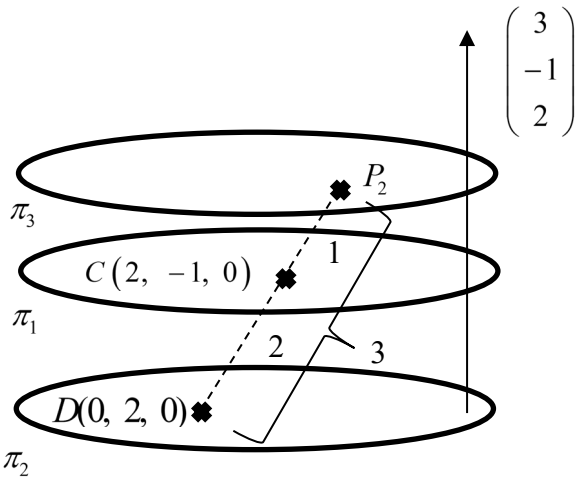
Case 2: π_1 is between π_2 and π_3
Let P_2 be a point on π_3 such that
$CD : CP_2 = 2 : 1$

Using ratio theorem,

$$\begin{pmatrix}2\\-1\\0\end{pmatrix}=\frac{2\overrightarrow{OP_2}+\begin{pmatrix}0\\2\\0\end{pmatrix}}{3}$$

$$\Rightarrow 2\overrightarrow{OP_2}=3\begin{pmatrix}2\\-1\\0\end{pmatrix}-\begin{pmatrix}0\\2\\0\end{pmatrix}$$

$$\Rightarrow \overrightarrow{OP_2}=\begin{pmatrix}3\\-\frac{5}{2}\\0\end{pmatrix}$$



Equation of π_3: $\mathbf{r}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=\begin{pmatrix}3\\-\frac{5}{2}\\0\end{pmatrix}\cdot\begin{pmatrix}3\\-1\\2\end{pmatrix}=9+\frac{5}{2}+0=\frac{23}{2}$

Cartesian equation of π_3: $3x-y+2z=\frac{23}{2}$

Source of Question: DHS CT2 9758/2023/01/Q11

7 Some computer games involve the locating of targets and planning actions on what to do with these targets.

In a particular computer-simulated scenario, a steep slope is modelled by the equation

$$p:\ \mathbf{r}=\begin{pmatrix}3\\1\\0\end{pmatrix}+\lambda\begin{pmatrix}2\\1\\0\end{pmatrix}+\mu\begin{pmatrix}4\\1\\2\end{pmatrix},$$

where λ and μ are real numbers.

(a) Express p in the form $\mathbf{r}.\begin{pmatrix}1\\a\\b\end{pmatrix}=c$, where the values of a, b, and c are to be determined.

[2]

(b) In this game, a first target is located on the slope such that it has the shortest distance to the point with coordinates (3, 8, 6). To destroy the target, a missile is set with a flight path given by the equation $l_1:\mathbf{r}=\begin{pmatrix}3\\8\\6\end{pmatrix}+\alpha\begin{pmatrix}h\\s\\t\end{pmatrix}$, where h, s and t are real values with s and t being positive, and α is a real parameter. (By setting the values of h, s and t, the direction of travel of the missile can be altered.)

(i) Determine the coordinates of the first target on the slope. [2]

(ii) Explain why it is not possible for the missile to approach the first target in the direction normal to the surface of the slope when h is set to be $s-t$. [2]

(iii) State a set of values of h, s and t such that makes it is possible for the missile to approach the first target in the direction normal to the surface of the slope. [1]

(c) A second target is located at position (3, 1, 0) on the slope. A surveillance drone is sent to capture close up pictures of the target. It is set to fly along the flight path given by equation $l_2 : \mathbf{r} = \begin{pmatrix} 3 \\ 8 \\ 6 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ m \\ 1 \end{pmatrix}$, where m is a constant, $m \neq 1$, and β is a real parameter. In order to capture the best view of the target, the drone should ideally get as close to the target as possible.

Given that the drone flies at an angle 60° to the slope, determine the coordinates of the position of the drone where it should stop to capture the best view of the target. [5]

Solution:

(a)	$p : \mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}$ $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -4 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$ $p : \mathbf{r}. \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}. \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = 1$ $a = -2,\ b = -1, c = 1$
(b)(i)	Since 1st target is the point on the slope nearest to (3,8,6), 1st target is the foot of perpendicular from (3,8,6) to slope. $\mathbf{r} = \begin{pmatrix} 3 \\ 8 \\ 6 \end{pmatrix} + \alpha \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3+\alpha \\ 8-2\alpha \\ 6-\alpha \end{pmatrix}$

	$\begin{pmatrix} 3+\alpha \\ 8-2\alpha \\ 6-\alpha \end{pmatrix}.\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}=1$ $3+\alpha-2(8-2\alpha)-(6-\alpha)=1$ $\alpha=\frac{10}{3}$ $\mathbf{r}=\begin{pmatrix} 3 \\ 8 \\ 6 \end{pmatrix}+\frac{10}{3}\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}=\frac{1}{3}\begin{pmatrix} 19 \\ 4 \\ 8 \end{pmatrix}$ Coordinates of first target is $\left(\frac{19}{3}, \frac{4}{3}, \frac{8}{3}\right)$.
(ii)	When $h=s-t,$ direction of the path is $\mathbf{d}=\begin{pmatrix} s-t \\ s \\ t \end{pmatrix}$ For the missile to have the shortest possible flight distance, it must take a perpendicular path to the point on the slope, ie, **d** // **n**, $\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}=k\begin{pmatrix} s-t \\ s \\ t \end{pmatrix}$ for some real k. Equating the x, y and z components, $1=k(s-t)\cdots(1)$ $-2=ks\cdots(2)$ $-1=kt\cdots(3)$ Solving (2) and (3), $ks-kt=-1$ RHS of (1) = $ks-kt=-1$ $\neq$ LHS of (1) Equations (1), (2) and (3) has no solution. Hence, it is not possible for the missile to hit the target in the shortest possible flight distance when h is set to be $s-t$.
(iii)	$\begin{pmatrix} h \\ s \\ t \end{pmatrix}//\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ with s and t positive. Set of values is $\{h,s,t\in\mathbb{R}: h=-n, s=2n, t=n \text{ where } n\in\mathbb{R}\}$

(c)

$$\sin 60^\circ = \frac{\left|\begin{pmatrix}1\\-2\\-1\end{pmatrix}.\begin{pmatrix}0\\m\\1\end{pmatrix}\right|}{\sqrt{6}\sqrt{m^2+1}} = \frac{|-2m-1|}{\sqrt{6}\sqrt{m^2+1}}$$

$$\frac{\sqrt{3}}{2} = \frac{|2m+1|}{\sqrt{6}\sqrt{m^2+1}}$$

$$\frac{3}{4} = \frac{4m^2+4m+1}{6\left(m^2+1\right)}$$

$$m^2-8m+7=0$$

$$(m-1)(m-7)=0$$

$$m=7 \quad (\text{Since given } m \neq 1)$$

Hence, $l_2 : \mathbf{r} = \begin{pmatrix}3\\8\\6\end{pmatrix} + \beta\begin{pmatrix}0\\7\\1\end{pmatrix}$.

$$\left[\left[\begin{pmatrix}3\\8\\6\end{pmatrix} + \beta\begin{pmatrix}0\\7\\1\end{pmatrix}\right] - \begin{pmatrix}3\\1\\0\end{pmatrix}\right] \bullet \begin{pmatrix}0\\7\\1\end{pmatrix} = 0$$

$$7(7+7\beta)+6+\beta=0$$

$$55+50\beta=0$$

$$\beta=-1.1$$

$$\mathbf{r} = \begin{pmatrix}3\\8\\6\end{pmatrix} - 1.1\begin{pmatrix}0\\7\\1\end{pmatrix} = \begin{pmatrix}3\\0.3\\4.9\end{pmatrix}$$

$\therefore$ Drone should stop at (3, 0.3, 4.9)

Source of Question: RVHS Prelim 9758/2025/P2/Q5

8 The Planetary Defense Coordination Office at NASA observes and tracks Near Earth Objects (NEOs) that could be potentially hazardous in a collision with Earth. Over short periods of time, vectors can be used to model the trajectories of the Earth and NEOs. They use coordinates (*x, y, z*) with units in millions of kilometers relative to the sun which is at position (0, 0, 0).

Earth's orbit is contained by the plane Π with equation $\mathbf{r} \bullet \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} = 0$.

(a) A satellite moves along a path with equation $\mathbf{r} = \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$, $\lambda \in \mathbb{R}$. Determine whether the satellite crosses Earth's orbital plane. [3]

Over a short period of time, Earth's motion can be modelled by the equation

$$\mathbf{r} = \begin{pmatrix} 15 \\ 20 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}, \mu \in \mathbb{R}.$$

An asteroid moves along a path with equation

$$\frac{x-10}{2} = y+2 = \frac{0.25-z}{k}$$

where k is a positive constant.

(b) Determine the possible values of k if the paths of the Earth and the asteroid do not intersect. [3]

It is now given that $k = 3$.

(c) Find the acute angle between the paths of the Earth and the asteroid. [2]

The asteroid is closest to the Earth when they are at points $(12, 18, -3)$ and $(12, -1, -2.75)$ on their respective paths.

(d) (i) Find the distance between the Earth and the asteroid at this instant. [2]

(ii) By considering the vector between the asteroid and the Earth at this instant, determine whether the distance found in part (d)(i) is the shortest distance between the two paths. [2]

Solution:

(a)

Since $\begin{pmatrix}1\\-1\\5\end{pmatrix}\bullet\begin{pmatrix}-2\\3\\1\end{pmatrix}=-2-3+5=0$, the satellite's path is parallel to Earth's Orbital plane.

Since $\begin{pmatrix}3\\2\\3\end{pmatrix}\bullet\begin{pmatrix}1\\-1\\5\end{pmatrix}=3-2+15=16\neq 0$, the satellite does not sit on Earth's Orbital plane.

Therefore, the satellite does not cross Earth's Orbital Plane.

Alternatively

Earth: $\mathbf{r}\bullet\begin{pmatrix}1\\-1\\5\end{pmatrix}=0---(1)$

Satellite: $\mathbf{r}=\begin{pmatrix}3\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}-2\\3\\1\end{pmatrix},\ \lambda\in\mathbb{R}---(2)$

Sub (2) into (1):

$\left[\begin{pmatrix}3\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}-2\\3\\1\end{pmatrix}\right]\bullet\begin{pmatrix}1\\-1\\5\end{pmatrix}=0---(3)$

$16+0\lambda=0$

There is no consistent value of λ for eqn (3)

Therefore, the satellite does not cross Earth's Orbital Plane.

(b)

$l_E:\mathbf{r}=\begin{pmatrix}15\\20\\-1\end{pmatrix}+\mu\begin{pmatrix}4\\-1\\-1\end{pmatrix},\ \mu\in\mathbb{R}$

$l_A:\dfrac{x-10}{2}=y+2=\dfrac{0.25-z}{k}$

$\mathbf{r}=\begin{pmatrix}10\\-2\\0.25\end{pmatrix}+t\begin{pmatrix}2\\1\\-k\end{pmatrix}$, where $t\in\mathbb{R}$

Consider

$\begin{pmatrix}10+2t\\-2+t\\0.25-tk\end{pmatrix}=\begin{pmatrix}15+4\mu\\20-\mu\\-1-\mu\end{pmatrix}$

	$2t-4\mu=5$ $t+\mu=22$ $\mu-tk=-1.25$ From GC, $t=15.5,\ \mu=6.5,\ tk=7.75$ $\Rightarrow k=\dfrac{7.75}{15.5}=0.5$ Therefore, the paths intersect when $k = 0.5$. Therefore, for the paths to not intersect, $k\neq 0.5,\ k\in\mathbb{R}^+$.
(c)	Let θ be the acute angle between the paths of the Earth and the asteroid. $\cos\theta=\left\|\begin{pmatrix}2\\1\\-3\end{pmatrix}\bullet\begin{pmatrix}4\\-1\\-1\end{pmatrix}\right\|$ $=\dfrac{\left\|8-1+3\right\|}{\sqrt{2^2+1^2+(-3)^2}\sqrt{4^2+(-1)^2+(-1)^2}}$ $=\dfrac{10}{\sqrt{14}\sqrt{18}}$ $\theta=\cos^{-1}\left(\dfrac{10}{\sqrt{14}\sqrt{18}}\right)=51.0°$ (1 d.p.)
(d) (i)	$\text{Distance}=\left\|\begin{pmatrix}12\\18\\-3\end{pmatrix}-\begin{pmatrix}12\\-1\\-2.75\end{pmatrix}\right\|$ $=\left\|\begin{pmatrix}0\\19\\-0.25\end{pmatrix}\right\|$ $=\sqrt{19^2+(-0.25)^2}$ $=19.0$ million km (3 s.f.)
(d) (ii)	The shortest distance between the paths will be when the vector connecting the two points on the paths is perpendicular to each of the two paths.

	$\begin{pmatrix} 0 \\ 19 \\ -0.25 \end{pmatrix} \bullet \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} = 19.75 \neq 0 \text{ – – – (1)}$ OR $\begin{pmatrix} 0 \\ 19 \\ -0.25 \end{pmatrix} \bullet \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix} = -18.75 \neq 0 \text{ – – – (2)}$ Since the vector is not perpendicular to one of the paths, the distance found is not the shortest distance between the paths. To show that $\left\lvert\begin{pmatrix} 0 \\ 19 \\ -0.25 \end{pmatrix}\right\rvert$ is not the shortest distance, it suffices to show either (1) or (2).

Source of Question: DHS Prelim 9758/2025/P1/Q9

9 The point A has coordinates $(1, -2, 4)$ and the plane π_1 has equation $2x - y + 2z = 5$.

(a) Find the exact shortest distance between A and π_1. [2]

The plane π_2 has equation $x + 3y - az = 3$, where a is a constant.

(b) Find the vector equation of line l, which is the line of intersection of π_1 and π_2, in terms of a. [4]

The plane π_3 has equation $bx - 2y + 4z = 3$, where b is a constant.

(c) Show that $(a-6)(b-4) = 0$ if l is parallel to π_3. [2]

(d) State the conditions that a and b must follow for the three planes to form a triangular prism, where all the planes are non-parallel and they do not have any point in common. Justify your answer. [4]

Solution:

(a)

$$\text{Shortest distance} = \frac{\left|5 - \begin{pmatrix}1\\-2\\4\end{pmatrix} \bullet \begin{pmatrix}2\\-1\\2\end{pmatrix}\right|}{\sqrt{2^2 + (-1)^2 + 2^2}}$$

$$= \frac{7}{3} \text{ units}$$

Alternative

By observation, (0,–5,0) is a point on the plane.

$$\text{Shortest distance} = \frac{\left|\left[\begin{pmatrix}0\\-5\\0\end{pmatrix} - \begin{pmatrix}1\\-2\\4\end{pmatrix}\right] \bullet \begin{pmatrix}2\\-1\\2\end{pmatrix}\right|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{7}{3} \text{ units}$$

Alternative

Let foot of perpendicular by F. Since F is on the line passing through (1,–2,4) and perpendicular to the π_1,

$$\overrightarrow{OF} = \begin{pmatrix}1\\-2\\4\end{pmatrix} + \mu \begin{pmatrix}2\\-1\\2\end{pmatrix}$$

Solving simultaneously,

$$\left[\begin{pmatrix}1\\-2\\4\end{pmatrix} + \mu \begin{pmatrix}2\\-1\\2\end{pmatrix}\right] \bullet \begin{pmatrix}2\\-1\\2\end{pmatrix} = 5$$

$$(2+2+8) + \mu(4+1+4) = 5$$

$$\mu = -\frac{7}{9}$$

$$\text{Shortest distance} = \left|\left[\begin{pmatrix}1\\-2\\4\end{pmatrix} - \frac{7}{9}\begin{pmatrix}2\\-1\\2\end{pmatrix}\right] - \begin{pmatrix}1\\-2\\4\end{pmatrix}\right|$$

$$= \left|-\frac{7}{9}\begin{pmatrix}2\\-1\\2\end{pmatrix}\right|$$

$$= \frac{7}{9}\sqrt{9}$$

$$= \frac{7}{3} \text{ units}$$

(b)

$2x - y + 2z = 5$

$x + 3y - az = 3$

Direction vector of l:

	$\begin{pmatrix}2\\-1\\2\end{pmatrix}\times\begin{pmatrix}1\\3\\-a\end{pmatrix}=\begin{pmatrix}a-6\\2a+2\\7\end{pmatrix}$ Let $z=0$, $2x-y=5---(1)$ $x+3y=3---(2)$ $x=\frac{18}{7}, y=\frac{1}{7}$ $\therefore l:\mathbf{r}=\begin{pmatrix}\frac{18}{7}\\\frac{1}{7}\\0\end{pmatrix}+\lambda\begin{pmatrix}a-6\\2a+2\\7\end{pmatrix}, \lambda\in\mathbb{R}$ **Alternative** $2x-y+2z=5---(1)$ $x+3y-az=3---(2)$ $2\times(2)-1: 7y-(2a+2)z=1$ $y=\frac{1}{7}+\frac{2a+2}{7}z$ $(2)+3\times(1): 7x+(-a+6)z=18$ $x=\frac{18}{7}+\frac{a-6}{7}z$ Let $z=\lambda'$, $l:\mathbf{r}=\begin{pmatrix}\frac{18}{7}\\\frac{1}{7}\\0\end{pmatrix}+\lambda'\begin{pmatrix}\frac{a-6}{7}\\\frac{2a+2}{7}\\1\end{pmatrix}, \lambda'\in\mathbb{R}$ $\therefore l:\mathbf{r}=\begin{pmatrix}\frac{18}{7}\\\frac{1}{7}\\0\end{pmatrix}+\lambda\begin{pmatrix}a-6\\2a+2\\7\end{pmatrix}, \lambda\in\mathbb{R}\ \ (\lambda=\frac{1}{7}\lambda')$
(c)	For l to be parallel to π_3, $\begin{pmatrix}a-6\\2a+2\\7\end{pmatrix}\bullet\begin{pmatrix}b\\-2\\4\end{pmatrix}=0$ $ab-6b-4a-4+28=0$ $ab-6b-4a+24=0$ $b(a-6)-4(a-6)=0$ $(a-6)(b-4)=0$ (shown)
(d)	$\pi_1:\mathbf{r}\bullet\begin{pmatrix}2\\-1\\2\end{pmatrix}=5$; $\pi_2:\mathbf{r}\bullet\begin{pmatrix}1\\3\\-a\end{pmatrix}=3$; $\pi_3:\mathbf{r}\bullet\begin{pmatrix}b\\-2\\4\end{pmatrix}=3$

For the three planes form a triangular prism,

(1) The line of intersection between π_1 and π_2, l, is parallel to $\pi_3 \Rightarrow a = 6$ **or** $b = 4$ (from **(c)**)

(2) l is not on π_3 regardless of b.

$$\begin{pmatrix} \frac{18}{7} \\ \frac{1}{7} \\ 0 \end{pmatrix} \bullet \begin{pmatrix} b \\ -2 \\ 4 \end{pmatrix} \neq 3 \; b\left(\tfrac{18}{7}\right) - 2\left(\tfrac{1}{7}\right) + 4(0) \neq 3 \Rightarrow b \neq \frac{23}{18}$$

(3) π_1 is not parallel to π_3

$$\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \neq m \begin{pmatrix} b \\ -2 \\ 4 \end{pmatrix} \Rightarrow m \neq \frac{1}{2} \Rightarrow b \neq 4$$

(4) π_2 is not parallel to π_3

$$\begin{pmatrix} 1 \\ 3 \\ -a \end{pmatrix} \neq k \begin{pmatrix} b \\ -2 \\ 4 \end{pmatrix} \Rightarrow k \neq -\frac{3}{2} \Rightarrow b \neq -\frac{2}{3} \text{ or } a \neq 6$$

$a = 6$ and $b \in \mathbb{R} \setminus \left\{-\frac{2}{3}, \frac{23}{8}, 4\right\}$